

ADS - Week - 2

L1.

JL Lemma - 4 - Towards Final Bound

n pts in $\mathbb{R}^d$ $\xrightarrow{\frac{1}{\sqrt{k}} R}$ $\mathbb{R}^k$ (with pairwise distances are preserved)
 $\nearrow k \times d \rightarrow N(0, 1)$

$$(1-\epsilon) \|x-y\|^2 \leq \|Mx - My\|^2 \leq (1+\epsilon) \|x-y\|^2 \quad \forall x, y \in S.$$

$\searrow \frac{1}{\sqrt{k}} R$

$$u \in \mathbb{R}^n \xrightarrow{\frac{1}{\sqrt{k}} R} v = \frac{1}{\sqrt{k}} R u \quad E(\|v\|^2) = \|u\|^2$$

$\rightarrow$ we came up with, $x = \frac{\sqrt{k} u}{\|u\|} \xrightarrow{\mathbb{R}^k}$
 $x \in \mathbb{R}^k \sim N(\vec{0}, I_{k \times k}) \xrightarrow{\mathbb{R}^k}$

$$P(\|v\|^2 \geq (1+\epsilon) \|u\|^2)$$

$$\Downarrow$$
$$P(\|x\|^2 \geq (1+\epsilon) k)$$

$\hookrightarrow$ about multivariate normal distribution

$$x \sim N(0, I) \Rightarrow [x_1, x_2, \dots, x_k]$$

$$E[\|x\|^2] = E\left[\sum_{i=1}^k x_i^2\right] = \sum_{i=1}^k E[x_i^2]$$

$$E[\|x\|^2] = k$$

$\text{var}(x_i)$

- Now we have to calculate the concentration bound,

$$P(\|x\|^2 \geq (1+\epsilon)k)$$

Markov
Chebyshev
so on } any will work

we will use, Chernoff's style bound

$$\Rightarrow P(e^{\lambda \|x\|^2} \geq e^{\lambda (1+\epsilon)k}) \quad \text{in built uses Markov for } \lambda > 0$$

$$P(x > k) \leq \frac{E[x]}{k}$$

$$\leq \frac{E[e^{\lambda \|x\|^2}]}{e^{\lambda (1+\epsilon)k}}$$

by Markov's Inequality

$$= \frac{E[e^{\lambda \sum_{i=1}^k x_i^2}]}{e^{\lambda (1+\epsilon)k}}$$

$$\|x\|^2 = \sum_{i=1}^k x_i^2$$

$$= \frac{E[e^{\lambda x_1^2 + \lambda x_2^2 + \dots}]}{e^{\lambda (1+\epsilon)k}}$$

$$= E\left[\prod_{i=1}^k e^{\lambda x_i^2}\right] / e^{\lambda (1+\epsilon)k}$$

$$= \prod_{i=1}^k E(e^{\lambda x_i^2}) / e^{\lambda (1+\epsilon)k}$$

this is a MGF

$$\|u\|^2 \leq (1+\epsilon)\|u\|^2 = \left(\prod_{i=1}^k \sqrt{\frac{1}{1-2\lambda}}\right) / e^{\lambda (1+\epsilon)k}$$

$$P(\text{bad event}) \leq \left(\frac{1}{1-2\lambda}\right)^{k/2} / e^{\lambda (1+\epsilon)k} \quad (\lambda > 0)$$

we want this bound to be as small as possible. One way could be we treat this as funcⁿ of λ differentiate w.r.t λ and equate it to 0.

$$= \exp \left(\log \left(\frac{1}{1-2\lambda} \right)^{k/2} - \lambda(1+\epsilon)k \right)$$

$$= \exp \left(k \left[\frac{1}{2} \log \left(\frac{1}{1-2\lambda} \right) - \lambda(1+\epsilon) \right] \right)$$

Taylor series expansion

$$\leq \exp \left(k \left(\lambda + 2\lambda^2 - \lambda - \lambda\epsilon \right) \right)$$

for log & small λ

$$P(\text{bad}) \leq \exp \left(k (2\lambda^2 - \lambda\epsilon) \right)$$

which choice of λ it will be least

$$F(\lambda) = 2\lambda^2 - \lambda\epsilon$$

$$F'(\lambda) \Rightarrow \lambda = \frac{\epsilon}{4}$$

$$F\left(\frac{\epsilon}{4}\right) = \frac{2\epsilon^2}{16} - \frac{\epsilon^2}{4}$$

$$= -\frac{\epsilon^2}{8}$$

for $\lambda = \frac{\epsilon}{4}$,

$$P(\text{bad}) \leq \exp \left(k \left(-\frac{\epsilon^2}{8} \right) \right)$$

L2

JL - 5 - Final Bound

$$P(\|u^2\|^2 \geq (1+\epsilon) \|u\|^2) = P(\|x^2\| \geq (1+\epsilon)k)$$

$$\leq e^{-\frac{k\epsilon^2}{8}}$$

So currently we have the upper bound only 1 pairwise distance vector. We need this to hold true for all the pairwise distances, and that too at the same time.

So now,

$$P\left(\exists \text{ a pair } (x_i, x_j) \text{ s.t. } \|Mx_i - Mx_j\|^2 \geq (1+\epsilon)\|x_i - x_j\|^2\right)$$

real bad event; that only 1 pair exists for which the information has not been preserved.

What will the complement of this event? that all the pairs (x_i, x_j) have their pairwise distances preserved.

$$= P\left(\bigcup_{\substack{(x_i, x_j) \\ nC_2 \text{ pairs}}} \|Mx_i - Mx_j\|^2 \geq (1+\epsilon)\|x_i - x_j\|^2\right)$$

all such pairs where my info. is not preserved.

as per Union bound

$$\leq \sum_{\substack{i, j \\ nC_2 \text{ pairs}}} P\left(\|Mx_i - Mx_j\|^2 \geq (1+\epsilon)\|x_i - x_j\|^2\right)$$

$$\leq n^2 e^{-\frac{\epsilon^2}{8}}$$

the chance of a bad event for any fixed vector is bounded by $e^{-\frac{\epsilon^2}{8}}$

$$P(\text{no pair sq. can deviates by more } (1+\epsilon)) \leq 0.001 \leftarrow \delta$$

$$P(\text{real bad event}) \leq n^2 e^{-k\epsilon^2/\delta} \leq \delta$$

$$\log(n^2) - \frac{k\epsilon^2}{\delta} \leq \log(\delta)$$

$$\Rightarrow \frac{k\epsilon^2}{\delta} \geq \log(n^2) - \log(\delta) = \log\left(\frac{n^2}{\delta}\right)$$

$$\boxed{k \geq \frac{\delta}{\epsilon^2} \log\left(\frac{n^2}{\delta}\right)}$$

see even this
final result doesn't
depend on 'd'

$$k \in O\left(\frac{\log(n/\delta)}{\epsilon^2}\right)$$

the original dimension

↳ complexity

- not only pairwise distances, even angles btw. the vectors is also preserved.

$$\boxed{P(\|v\|^2 \leq (1-\epsilon)\|u\|^2) \leq e^{-\frac{k\epsilon^2}{\delta}}}$$

↳ the similar analysis goes on here as well.

$$P\left(\begin{array}{l} \|u\|^2 \leq (1-\epsilon)\|v\|^2 \text{ or} \\ \|v\|^2 \geq (1-\epsilon)\|u\|^2 \end{array}\right) \leq 2e^{-\frac{k\epsilon^2}{\delta}}$$

$$P(\text{real bad event}) \leq 2n^2 e^{-k\epsilon^2/\delta} \leq \delta$$

$$k \geq \frac{\delta}{\epsilon^2} \log\left(\frac{2n^2}{\delta}\right)$$

L3

TL: Linear Algebra Interpretation

- we saw this problem as estimation & not actually transformation from d to k .

$$M = \frac{1}{\sqrt{k}} R$$

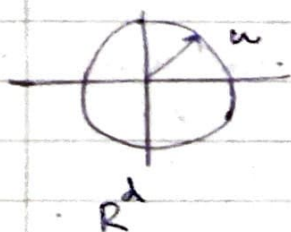
$$M_{k \times d} = \underbrace{U}_{k \times k} \underbrace{\Sigma}_{k \times d} \underbrace{V^T}_{d \times d}$$

orthonormal matrix $k \times d$ pseudo diagonal matrix orthonormal matrix

$$Mu = U \Sigma V^T u$$

singular values

$$\begin{bmatrix} \delta_1 & & & 0 \\ & \delta_2 & & \\ & & \ddots & \\ 0 & & & \end{bmatrix}$$



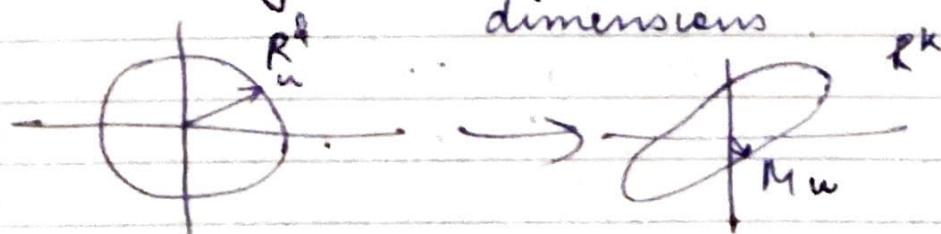
$$Mu = \underbrace{U}_{k \times k} \underbrace{\Sigma}_{k \times d} \underbrace{V^T}_{d \times d} u$$

$d \times 1$

firstly, the u get isolated in the original dimension

now I am scaling into k dimensions

again doing the isolation in k dimensions



- when we do, $v^T u$, it is a random orthogonal transformation (random rotation). It will end up being rotated anywhere uniformly.

$$\sum v^T u = \sum_{i=1}^n \sigma_i a_i \rightarrow \approx \frac{1}{\sqrt{n}}$$

singular value will look sth. like $\approx \sqrt{\frac{n}{k}}$

it will always lie in

the interval of $\left(\sqrt{\frac{n}{k}} - 1, \sqrt{\frac{n}{k}} + 1\right)$

$$= \begin{bmatrix} \frac{1}{\sqrt{k}} \\ \vdots \end{bmatrix}$$

$$v = U \Sigma V^T u = \approx \begin{bmatrix} \frac{1}{\sqrt{k}} \\ \vdots \end{bmatrix}$$

↳ random uniform rotation in $\mathbb{R}^k$

No single direction is preferred.

$$\|v\|^2 = 1$$

These things will work for any value of k .

But it works in high dim.

d must be very very large.

- when d is small, it has bias towards some direction.